

Visualizing the cost function

Model:

$$f_{w,b} = wx + b$$

Params:

w, b

(Previously we did $b=0$)
(assume)

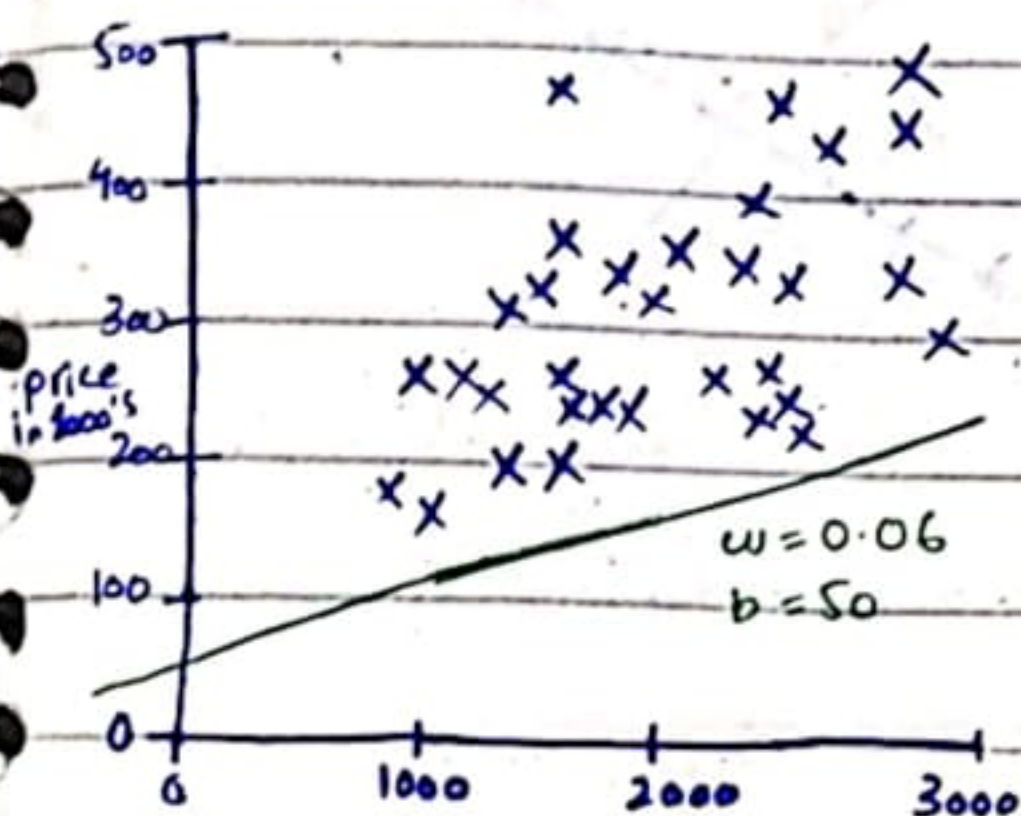
Cost Function:

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Objective:

minimize $J(w,b)$
 w, b

$f_{w,b}$ (func of x)

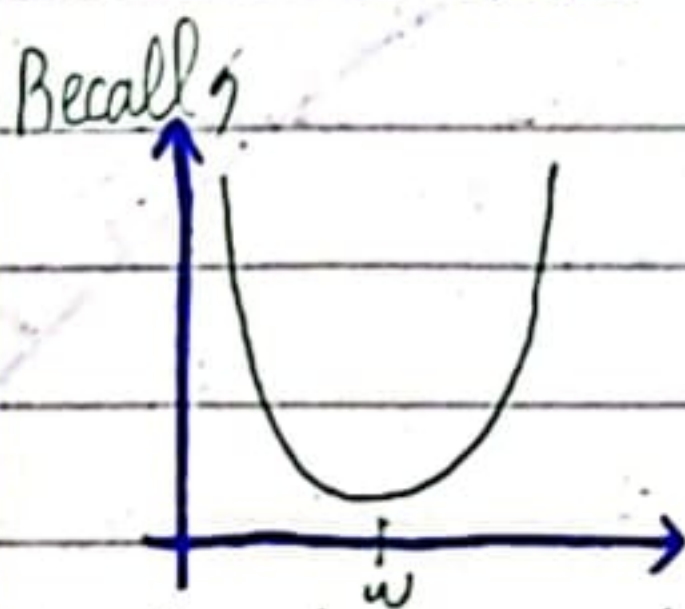


size in ft^2

$$f_{w,b}(x) = 0.06x + 50$$

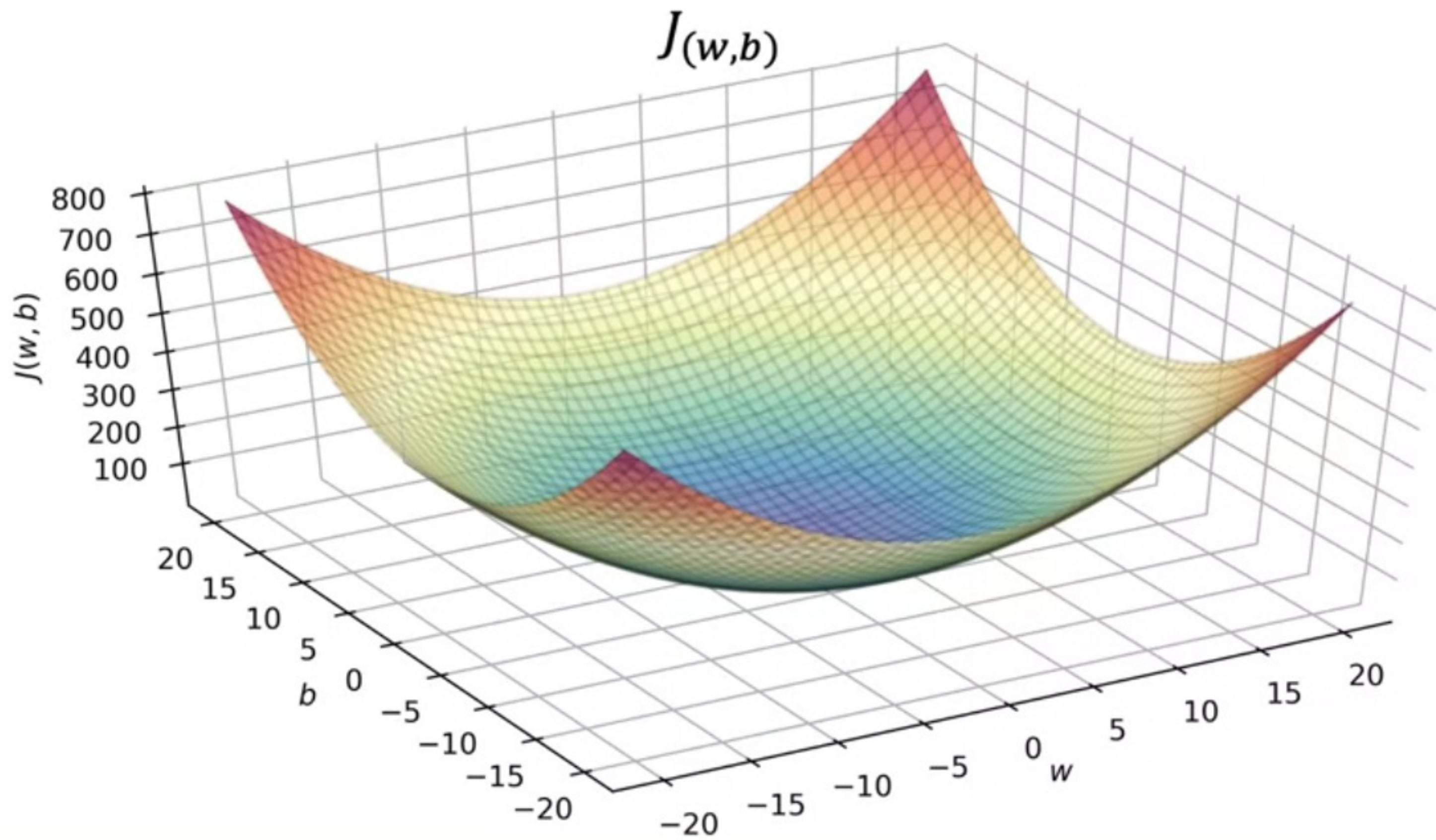
→ pretty bad model for this set as you can see (underestimating housing prices).

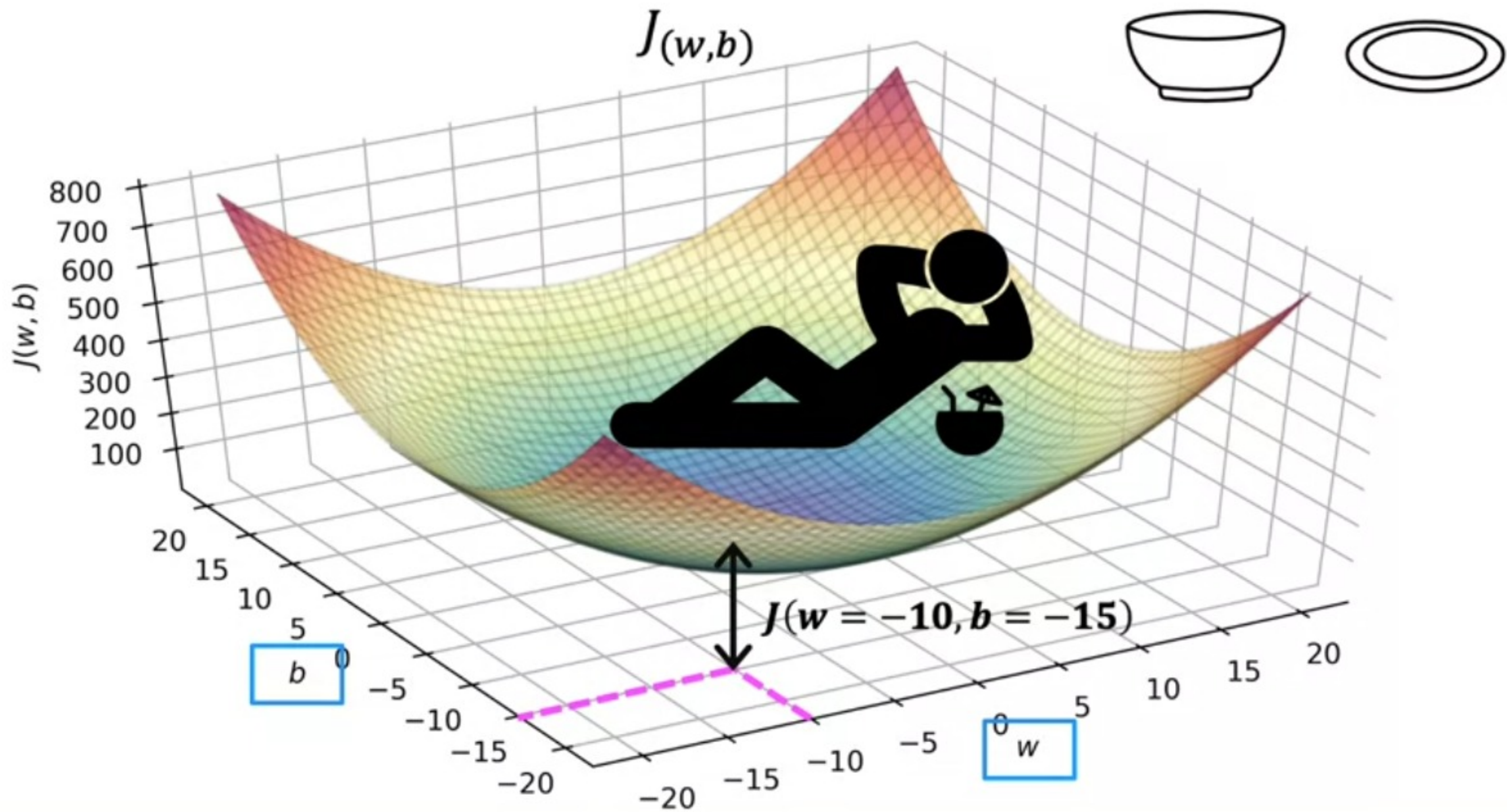
J (func of w, b)

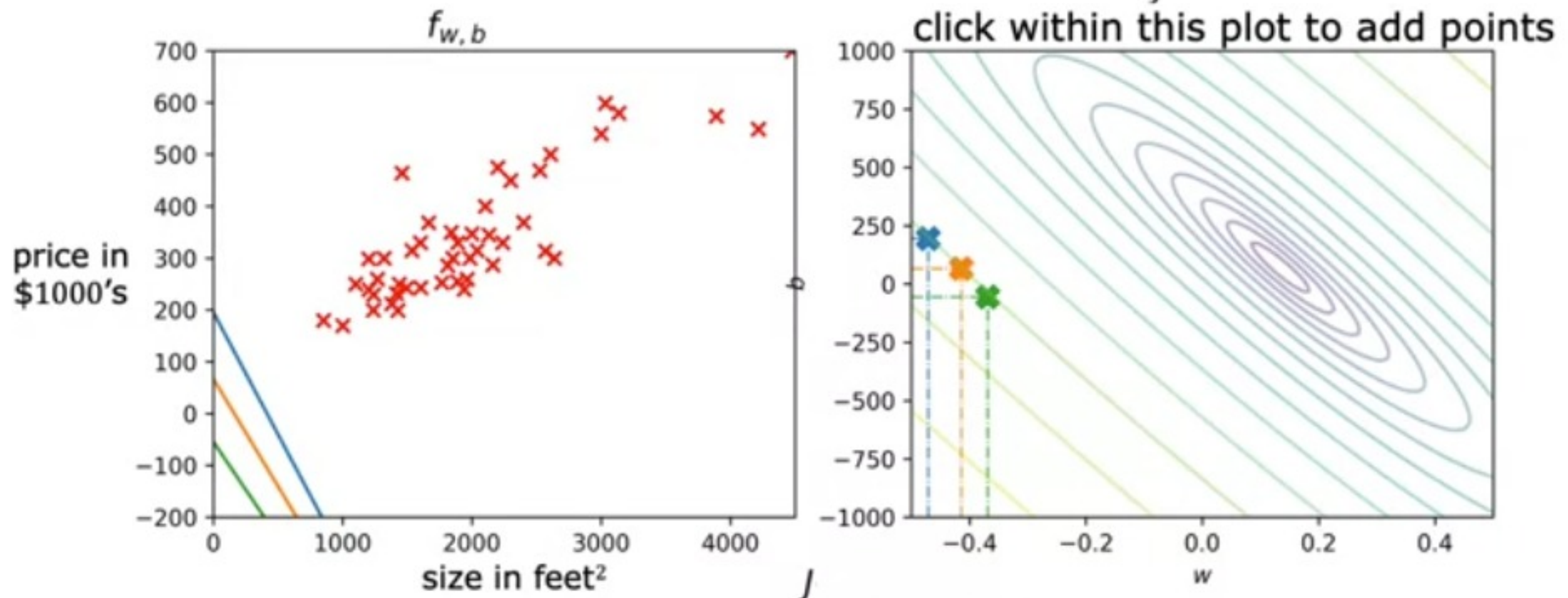


• recall when we only had w , the c.f had this U-shaped curve.
• Now, we have w, b so plot becomes complex.

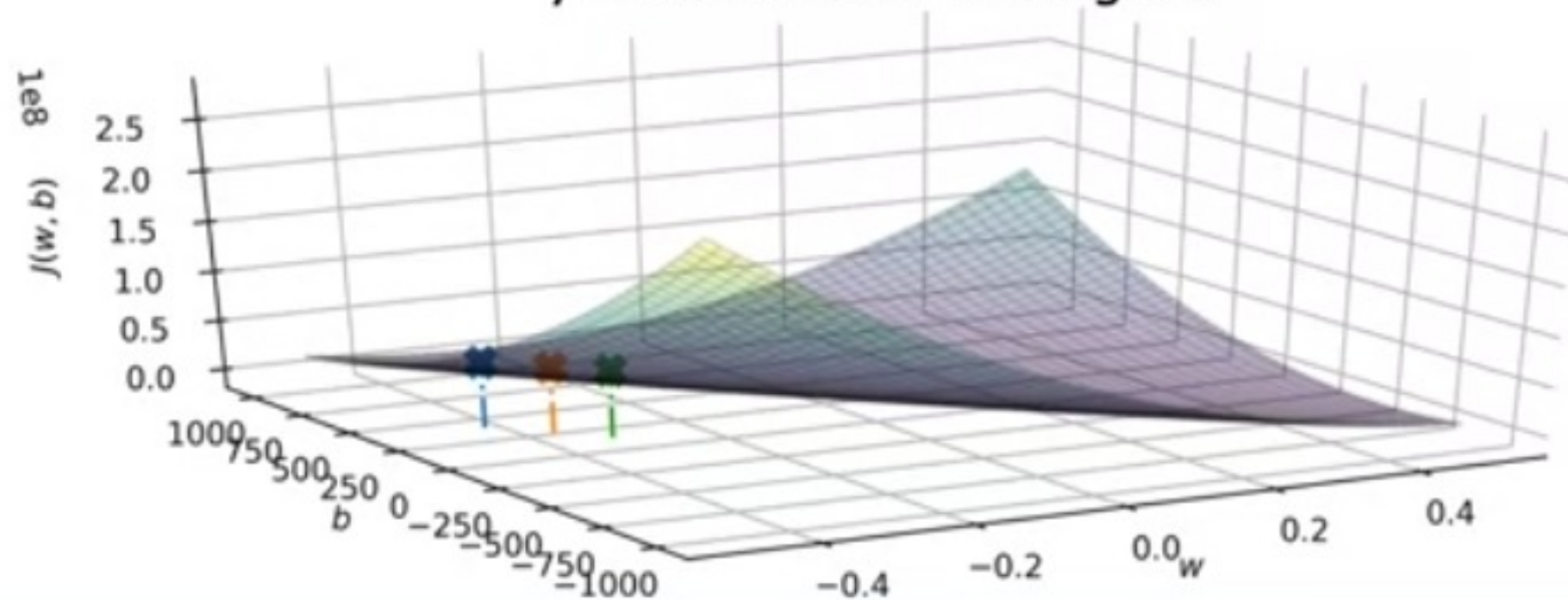
• for two params, we still have bowl shaped of the c.f as shown next.







you can rotate this figure



Bottom :

The bottom graph is the actual 3D bowl for w & b parameters.

- x -axis = w
- y -axis = b
- height(z -axis) = J

Top Right :

A contour plot is basically a bird's eye view (top down) of the same 3D bowl, but instead of drawing the whole 3D shape, it draws rings/loops that connects all the points that have same height (value J).

- Each ring = "all (w, b) combination that give this exact J ." ^{same J}
- Rings close together = steep slope _(same badness of fit)
- The very center of the smallest, inner most ring = the bottom of bowl = the best w, b (lowest J).

As you move outward, each bigger ring represents higher, worse J value.

Top left:

Just shows how line looks on the real data as you can see All three are bad fits they barely touch the data. That why their dots in both graphs are far away from the center, and have same J even though w, b are diff as horizontal slicing (the same badness concept).

Conclusion:

So basically, we are trying to find combinations of w, b that are close to center of rings = bottom of bowl = matches your data with.